A Graph-Theoretic Kun Repair Theorem

Sparse-set sharpening, resolvent smoothing, and expander decomposition

Expository note

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Status of the quantitative statement. Kun's theorem says qualitatively that finite graphs locally converging to the Cayley graph of a property-(T) group are asymptotically editable into disjoint unions of uniform expanders. The graph-theoretic theorem below is a streamlined quantitative formulation derived from the same mechanism, with a resolvent smoothing step replacing fixed-time random-walk localization. In particular, the explicit $O_\gamma(\varepsilon^{1/3})$ edit bound below is presented as a derivation here, not as a quoted numerical bound from Kun's paper.

1 The point of the theorem

Let G be a finite k-regular graph, with Markov operator K and normalized Laplacian

$$L = I - K.$$

All L_2 norms and inner products are with respect to the uniform probability measure on $V(G)$. Thus

$$\langle f, g\rangle = \mathbb{E}_{x\in V} f(x)g(x), \qquad \mathcal{E}[f] = \langle f, Lf\rangle = \frac{1}{2}\mathbb{E}_{x\sim y}(f(x) - f(y))^2.$$

For $T \subseteq V$ put $\mathrm{vol}(T) = |T|/|V|$. With $\partial_G T$ denoting the directed generator-edge boundary,

$$\mathcal{E}[\mathbf{1}_T] = \frac{|\partial_G T|}{k|V|}, \qquad \Phi_G(T) = \frac{\mathcal{E}[\mathbf{1}_T]}{\mathrm{vol}(T)} = \frac{|\partial_G T|}{k|T|}.$$

For a genuine finite action of a group having an SOS certificate

$$\Delta^2 - \gamma\Delta = \sum_j \xi_j^* \xi_j,$$

the nonzero spectrum of the normalized Laplacian is at least γ. In an approximate action, however, the certificate is only locally valid. The key observation is that all group theory can be compressed into one graph-theoretic hypothesis: $L^2 - \gamma L$ is positive semidefinite on functions avoiding a small exceptional set.

2 The group-free repair theorem

We may assume $0 < \gamma \le 1$; replacing a larger γ by $\min\{\gamma, 1\}$ only weakens the hypothesis.

Theorem 2.1 (Quantitative repair from an almost-everywhere spectral certificate)**.** *Let G be a finite k-regular graph on n vertices. Suppose there is a set $B \subseteq V(G)$, $|B| \le \varepsilon n$, such that*

$$\|Lf\|_2^2 \ge \gamma\, \mathcal{E}[f] \qquad \text{for every } f \text{ satisfying } f|_B = 0. \tag{1}$$

We refer to this as hypothesis (H)*. Then there is a partition*

$$V = Z \sqcup C_1 \sqcup \cdots \sqcup C_m$$

such that

$$|Z| + \frac{1}{k}\sum_i |\partial_G C_i| \le C\,\gamma^{-4/3}\varepsilon^{1/3} n, \tag{2}$$

and every C_i has strong internal expansion:

$$|E_G(A, C_i \setminus A)| \ge c\gamma k|A| \qquad (A \subseteq C_i,\ |A| \le |C_i|/2)\,. \tag{3}$$

Here $c, C > 0$ are universal constants.

Consequently one may make the vertices of Z singletons and repair the missing transition slots inside each C_i, producing a k-regular graph G' on the same vertex set whose nonsingleton connected components satisfy

$$\Phi(G'[C_i]) \ge c\gamma, \qquad \lambda_1(L_{G'[C_i]}) \ge c'\gamma^2.$$

If G is an S-labelled permutation graph, the repair may be performed within the class of S-labelled permutation graphs, changing at most

$$C\gamma^{-4/3}\varepsilon^{1/3}$$

of all generator values, up to a constant depending only on the harmless choice of directed-versus-undirected edge normalization.

The proof has four steps:

$$\boxed{\text{local PSD} \Longrightarrow \text{small diagonal defect} \Longrightarrow \text{sparse-set sharpening} \Longrightarrow \text{peeling + trimming.}}$$

Only the first implication remembers anything about the origin of the hypothesis.

3 From an exceptional set to a diagonal error

Put

$$A = L^2 - \gamma L$$

and let P be multiplication by $\mathbf{1}_{V\setminus B}$. Hypothesis (1) is exactly

$$PAP \succeq 0.$$

Set $F = A - PAP$. Every nonzero entry of F has at least one endpoint in B. Define

$$b(x) = \sum_y |F_{xy}|.$$

Since F is symmetric and $2|uv| \le u^2 + v^2$,

$$\langle f, Ff \rangle \ge -\mathbb{E}[bf^2].$$

Hence

$$\boxed{\|Lf\|_2^2 \ge \gamma \mathcal{E}[f] - \mathbb{E}[bf^2].} \tag{4}$$

Moreover the total defect mass is $O(\varepsilon)$. Indeed,

$$A = (1-\gamma)I + (\gamma - 2)K + K^2,$$

so every absolute row sum of A is at most 4. Since F consists only of entries incident to B,

$$\boxed{\beta := \mathbb{E}b \le 8\varepsilon.} \tag{5}$$

This is the first conceptual simplification: instead of deleting a radius-t neighborhood of every bad vertex, we pay an additive defect whose *total mass* is $O(\varepsilon)$.

4 Resolvent smoothing

For $t > 0$ define the resolvent

$$Q_t = (I + tL)^{-1}.$$

Since $L = I - K$,

$$Q_t = \frac{1}{1+t} \sum_{j=0}^{\infty} \left(\frac{t}{1+t}\right)^j K^j. \tag{6}$$

Thus Q_t is averaging by a geometrically stopped random walk of mean t. In particular Q_t is self-adjoint, positivity-preserving, and mean-preserving.

Lemma 4.1 (Smoothing while staying near the original function)**.** *Assume* (4)*. For any u, put $f = Q_t u$. Then*

$$\|f - u\|_2^2 \le \frac{\mathcal{E}[u]}{\gamma} + t^2 \mathbb{E}[bf^2], \tag{7}$$

$$\mathcal{E}[f] \le \frac{\mathcal{E}[u]}{\gamma^2 t^2} + \frac{2}{\gamma} \mathbb{E}[bf^2]. \tag{8}$$

Proof. For every $x \ge 0$,

$$\left(\frac{tx}{1+tx}\right)^2 \le \frac{x}{\gamma} + t^2 \frac{\gamma x - x^2}{(1+tx)^2}. \tag{9}$$

Indeed, after multiplying the difference between the right and left sides by $\gamma(1+tx)^2/x$ (with $x = 0$ trivial), one obtains

$$1 + 2tx + t^2(x-\gamma)^2 \ge 0.$$

Applying the spectral theorem to L, and using $f = (I + tL)^{-1}u$, gives

$$\|u - f\|_2^2 \le \frac{\mathcal{E}[u]}{\gamma} + t^2 \langle f, (\gamma L - L^2) f \rangle.$$

By (4), the final inner product is at most $\mathbb{E}[bf^2]$, proving (7).

Also $(I + tL)f = u$, hence

$$Lf = \frac{u - f}{t}.$$

Using (4) once more,

$$\gamma \mathcal{E}[f] \le \frac{\|u - f\|_2^2}{t^2} + \mathbb{E}[bf^2],$$

and substituting (7) gives (8). □

When $b = 0$ and $t \to \infty$, Q_t converges to projection onto $\ker L$, and (7) becomes the usual spectral-gap estimate

$$\operatorname{dist}(u, \ker L)^2 \le \frac{\mathcal{E}[u]}{\gamma}.$$

Thus the lemma is best viewed as a robust, finite-scale version of projection onto invariant functions.

5 Sparse-set sharpening

Fix a sufficiently small universal constant $c_0 > 0$ (say $c_0 = 10^{-3}$) and put

$$h = c_0 \gamma.$$

Smooth the defect density itself:

$$b_t = Q_t b,$$

and declare a vertex bad at scale t if

$$b_t(x) > \frac{c_0}{t^2}.$$

Thus

$$B_t = \left\{ x : b_t(x) > c_0/t^2 \right\}.$$

Since Q_t preserves averages, Markov's inequality and (5) give

$$\operatorname{vol}(B_t) \le c_0^{-1} \beta t^2. \tag{10}$$

Now let

$$\varnothing \ne T \subseteq V \setminus B_t, \qquad \Phi_G(T) < h.$$

Set

$$u = \mathbf{1}_T, \qquad f = Q_t \mathbf{1}_T.$$

Because $0 \le f \le 1$ and Q_t is self-adjoint,

$$\begin{aligned} \mathbb{E}[bf^2] \le \mathbb{E}[bf] &= \langle b, Q_t \mathbf{1}_T \rangle = \langle Q_t b, \mathbf{1}_T \rangle \\ &\le \frac{c_0}{t^2} \operatorname{vol}(T). \end{aligned}$$

Also $\mathcal{E}[u] < c_0 \gamma \operatorname{vol}(T)$. Lemma 4.1 therefore yields

$$\|f - \mathbf{1}_T\|_2^2 \le 2c_0 \operatorname{vol}(T), \tag{11}$$

$$\mathcal{E}[f] \le \frac{3c_0}{\gamma t^2} \operatorname{vol}(T). \tag{12}$$

For $\theta \in [1/3, 2/3]$ put

$$U_\theta = \{x : f(x) > \theta\}.$$

Every point of $U_\theta \triangle T$ contributes at least $1/9$ to (11), so every such threshold satisfies

$$|U_\theta \triangle T| \le 18 c_0 |T| < \frac{1}{10}|T|. \tag{13}$$

It remains to choose a threshold of small boundary. This is precisely the coarea/Cauchy–Schwarz sweep used in the usual proof of Cheeger's inequality. Since $\mathbb{E} f = \mathrm{vol}(T)$,

$$\Pr[f(x) > 1/3] \le 3\,\mathrm{vol}(T).$$

For a stationary random directed edge (x, y),

$$\Pr[\max(f(x), f(y)) > 1/3] \le 6\,\mathrm{vol}(T).$$

Consequently

$$\begin{aligned} 2\int_{1/3}^{2/3} \mathcal{E}[\mathbf{1}_{U_\theta}]\, d\theta &\le \mathbb{E}_{x\sim y}\left[|f(x) - f(y)|\, \mathbf{1}_{\{\max(f(x),f(y))>1/3\}}\right] \\ &\le \sqrt{12\,\mathrm{vol}(T)\mathcal{E}[f]}. \end{aligned}$$

Using (12) and the fact that the threshold interval has length $1/3$, some $\theta \in [1/3, 2/3]$ satisfies

$$\boxed{|U_\theta \triangle T| \le \frac{1}{10}|T|, \qquad \Phi_G(U_\theta) \le \frac{1}{t\sqrt{\gamma}}.} \tag{14}$$

(The numerical choice of c_0 only serves to absorb the universal constants.)

Thus we have proved the central analytic statement.

Proposition 5.1 (Sparse-set sharpening)**.** *Outside a set of volume $O(\beta t^2)$, every set of conductance less than the fixed constant $h = c_0\gamma$ is within $|T|/10$ vertices of a set of conductance at most*

$$\alpha = \frac{1}{t\sqrt{\gamma}}.$$

The striking point is that h is fixed while α can be made arbitrarily small by increasing t; the price is only an exceptional mass $O(\beta t^2)$, not an exponentially growing neighborhood of the original defects.

6 Minimum-set peeling

We now forget how Proposition 5.1 was proved. Start with

$$R = V \setminus B_t.$$

As long as the current remainder contains a nonempty set $T \subseteq R$ with

$$\Phi_G(T) < h,$$

choose one of minimum cardinality, sharpen it to a set U satisfying (14), and peel off

$$P = U \cap R.$$

When no such T remains, keep the terminal remainder as one final piece.

Every peeled piece is a near-expander

Since $|U \triangle T| \le |T|/10$,

$$0.9|T| \le |P| \le 1.1|T| < 2|T|.$$

Hence if $A \subseteq P$ and $|A| \le |P|/2$, then $|A| < |T|$. Since A lies in the current remainder, the minimality of T gives

$$\boxed{|\partial_G A| \ge hk|A|.} \tag{15}$$

The terminal remainder has the same property by the stopping rule.

The boundary in (15) is the boundary in the *original* graph. Thus a piece is not yet an induced expander: some edges witnessing its expansion may leave the piece.

Only few edges cross between the pieces

The sets P are disjoint. Moreover

$$|U| \le 1.1|T| \le \frac{11}{9}|P|,$$

so

$$\sum_{\text{peels}} |U| = O(n).$$

Every edge between two distinct final pieces is charged to the boundary of the earlier sharpened set U; edges incident to B_t are charged to B_t. Writing

$$b_i = |\partial_G P_i|,$$

we obtain

$$\boxed{\sum_i b_i = O\big(kn\big(\mathrm{vol}(B_t) + \alpha\big)\big)\,.} \tag{16}$$

There is no logarithmic loss: every crossing edge is charged at its first separation, while the total size of the sharpened sets is only $O(n)$.

7 Trimming near-expanders to induced expanders

The following elementary lemma is the existential analogue of the "trimming" or "pruning" operations that play a central role in modern expander-decomposition algorithms.

Lemma 7.1 (Trimming)**.** *Suppose $P \subseteq V$ satisfies*

$$|\partial_G A| \ge hk|A| \qquad (A \subseteq P,\ |A| \le |P|/2),$$

and

$$b := |\partial_G P| \le \frac{hk|P|}{20}.$$

Then there is $C \subseteq P$ such that

$$|P \setminus C| \le \frac{2b}{hk}, \tag{17}$$

$$|\partial_G C| \le 2b, \tag{18}$$

and

$$|E_G(A, C \setminus A)| \geq \frac{hk}{2}|A| \qquad (A \subseteq C,\ |A| \leq |C|/2). \tag{19}$$

Proof. Start with $C = P$. Whenever there is a nonempty $A \subseteq C$, $|A| \leq |C|/2$, violating (19), remove A from C. Let $D = P \setminus C$ be everything removed so far. Summing the sparse cuts at the moments of removal gives

$$|E_G(D, C)| \leq \frac{hk}{2}|D|. \tag{20}$$

As long as $|D| \leq |P|/2$, the near-expansion of D gives

$$hk|D| \leq |E_G(D, C)| + b \leq \frac{hk}{2}|D| + b,$$

so

$$|D| \leq \frac{2b}{hk} \leq \frac{|P|}{10}. \tag{21}$$

The process cannot jump past $|P|/2$. At a first such jump, before the final removal one has $|D| \leq |P|/10$, and the rule removes at most half of the current C; hence after the jump

$$\frac{9|P|}{20} \leq |C| < \frac{|P|}{2}.$$

Applying near-expansion to this C, and using (20), gives

$$hk|C| \leq \frac{hk}{2}(|P| - |C|) + b,$$

therefore

$$|C| \leq \frac{|P|}{3} + \frac{2b}{3hk} \leq \frac{11|P|}{30},$$

a contradiction. Thus (21) holds throughout. At termination (19) holds, while

$$|\partial_G C| \leq b + |E_G(D, C)| \leq 2b.$$

□

Apply the lemma to every piece with

$$b_i \leq hk|P_i|/20.$$

Discard the remaining pieces entirely. By (16), the total mass of discarded pieces is at most

$$\frac{20}{hk}\sum_i b_i.$$

For each retained piece, trim to a core C_i and add the removed vertices to Z. Equations (17)–(18) and (16) yield

$$\frac{|Z|}{n} + \frac{1}{kn}\sum_i |\partial_G C_i| \leq \frac{C}{\gamma}\left(\beta t^2 + \frac{1}{t\sqrt{\gamma}}\right). \tag{22}$$

Every retained core has internal conductance at least $h/2 = \Omega(\gamma)$.

Now choose

$$t = (\beta\sqrt{\gamma})^{-1/3}$$

when $\beta > 0$. Then the two terms in (22) balance, and

$$\boxed{\frac{|Z|}{n} + \frac{1}{kn}\sum_i |\partial_G C_i| \le C\gamma^{-4/3}\beta^{1/3} \le C'\gamma^{-4/3}\varepsilon^{1/3}.} \tag{23}$$

When $\varepsilon = 0$, the hypothesis is global and the graph is already a disjoint union of components with nonzero Laplacian spectrum at least γ.

This proves the partition part of Theorem 2.1.

8 Restoring regularity and labels

Make each vertex of Z a singleton. Inside each core C_i, retain every original transition whose two endpoints both lie in C_i.

For an S-labelled graph, fix a generator s. The retained s-arrows form a partial bijection $C_i \dashrightarrow C_i$. The number of missing sources equals the number of missing targets, so complete this partial bijection arbitrarily to a permutation of C_i. Choose the completion for s^{-1} to be the inverse of that for s. If $s = s^{-1}$, extend the partial involution to an involution (fixed points are allowed).

Only transitions that crossed the final partition need to be changed. Their total number is controlled by (23). Moreover every original edge internal to C_i has been retained, so the completion cannot decrease the internal expansion from Lemma 7.1. Therefore

$$\Phi(G'[C_i]) = \Omega(\gamma).$$

By the easy direction of Cheeger's inequality in the normalization used here,

$$\Phi(H) \le 2\sqrt{\lambda_1(L_H)},$$

so

$$\lambda_1(L_{G'[C_i]}) = \Omega(\gamma^2).$$

Theorem 2.1 follows.

9 The SOS/Kazhdan corollary: minimal group assumptions

Now let

$$\Gamma = \langle S \rangle, \qquad S = S^{-1}, \qquad |S| = k,$$

and suppose that in $\mathbb{R}[\Gamma]$ we are given a finite-support SOS identity

$$\boxed{\Delta^2 - \gamma\Delta = \sum_{j=1}^{m} \xi_j^* \xi_j.} \tag{24}$$

Ozawa proved that the existence of such a certificate for some $\gamma > 0$ is equivalent to property (T) for a finitely generated group [2]. For the repair theorem, however, we need only the displayed finite identity itself.

Choose word representatives for all group elements occurring in the ξ_j. Expanding (24) involves only finitely many equalities between bounded-length words. Call a vertex of an S-graph *certificate-correct* if all those finitely many word equalities hold when based at that vertex.

Corollary 9.1 (Fixed-radius Kun repair from an SOS certificate)**.** *Fix the finite-support certificate* (24)*. There is a finite radius R_0 (indeed $R_0 = O(D)$ if the certificate has word degree D) such that the following holds.*

If a finite S-graph G has at least $(1-\varepsilon)|V(G)|$ certificate-correct vertices – in particular, if at least that many vertices are R_0-consistent with Γ – then G can be repaired, by changing an

$$O_\gamma(\varepsilon^{1/3})$$

fraction of generator values, into an S-labelled graph whose connected components are singletons or have conductance $\Omega(\gamma)$ and normalized Laplacian gap $\Omega(\gamma^2)$.

Proof. Substitute the generator permutations of G into the chosen word representatives in (24). Let Ξ_j be the resulting finite-graph operators. At every certificate-correct vertex, the expanded matrix identity

$$L^2 - \gamma L = \sum_j \Xi_j^* \Xi_j$$

holds on the corresponding basis vector. If B is the set of incorrect vertices and $f|_B = 0$, self-adjointness gives

$$\langle f, (L^2 - \gamma L) f \rangle = \sum_j \|\Xi_j f\|_2^2 \geq 0.$$

Thus hypothesis (1) of Theorem 2.1 holds. Apply that theorem. □

This formulation removes several hypotheses that are often bundled into the group-theoretic statement:

- **Finite presentability is unnecessary.** Only the finitely many word identities appearing in the fixed SOS certificate matter.
- **Faithfulness is unnecessary.** No nonrelation needs to be detected; local consistency suffices.
- **Full Cayley-ball correctness is unnecessary.** It is enough that the finite list of certificate identities hold.
- **Property (T) is not an extra hypothesis once the SOS is given.** Its only role here is to guarantee the existence of some finite-support certificate.

After passing to hypothesis (1), all group theory disappears.

10 A converse: the graph-theoretic hypothesis is essentially exact

There is a useful converse explaining why (1) is the right abstraction. Suppose G is edited to a graph G' which is a disjoint union of components having normalized Laplacian gaps at least $\rho > 0$. Let B contain every endpoint of an edited edge. If $f|_B = 0$, then the two graphs act identically on f along every edge touched by the quadratic forms, so

$$L_G f = L_{G'} f, \qquad \mathcal{E}_G[f] = \mathcal{E}_{G'}[f].$$

Since

$$L_{G'}^2 \succeq \rho L_{G'},$$

we get

$$\|L_G f\|_2^2 \geq \rho \mathcal{E}_G[f].$$

Thus, for bounded-degree graph sequences, the qualitative equivalence is

$$\boxed{\begin{array}{c} o(n)\text{-close to a union of uniform expanders} \\ \Longleftrightarrow \\ L^2 - \gamma L \text{ is PSD on functions avoiding an } o(n) \text{ set, for some fixed } \gamma > 0. \end{array}}$$

This is the graph-theoretic core behind Kun's characterization by the Markov operator [1].

11 Relation to modern expander decomposition and sparsification

11.1 Why this is stronger than generic expander decomposition

A standard (ϕ, η) expander decomposition partitions V into clusters whose induced subgraphs have conductance at least ϕ, while at most an η fraction of edges cross between clusters. Modern algorithms seek small η as a function of ϕ and n; for example, Agassy–Dorfman–Kaplan obtain roughly $\eta = \phi \log^2 n$ in nearly linear time [4].

The property-(T)/Kun situation is structurally stronger. Here the two parameters are decoupled:

$$\boxed{\text{internal conductance } \Omega(\gamma) \quad \text{and} \quad \text{crossing/edit fraction } O_\gamma(\varepsilon^{1/3}).}$$

For fixed γ, the crossing fraction tends to zero with the local defect ε, while the component expansion stays bounded below by a constant. Generic expander decomposition cannot promise this for arbitrary graphs; the sparse-set-sharpening property is the extra rigidity.

The proof is existential rather than algorithmic: the minimum-cardinality sparse set in the peeling step need not be efficiently findable. The resolvent and sweep parts are algorithmically familiar, but the selection rule is deliberately optimized for conceptual simplicity rather than running time.

11.2 Trimming and pruning

The passage from (15) to induced expander cores is exactly the role played by *trimming* or *expander pruning* in modern decomposition algorithms. Saranurak and Wang developed efficient versions that, after a controlled number of boundary-edge deletions, remove only controlled volume while retaining a constant-factor fraction of the original conductance [3]. Lemma 7.1 is a short existence-level analogue tailored to this proof.

11.3 Resolvents, PageRank, and spectral partitioning

The operator

$$Q_t = (I + tL)^{-1}$$

is a geometrically stopped random walk, and is closely related to personalized PageRank-type smoothing. Local partitioning by PageRank vectors similarly combines a diffusion operator with a threshold sweep to reveal sparse cuts [5]. Here the direction of use is somewhat different: we already know the set T, and the spectral certificate says that resolvent smoothing can make its energy tiny *without moving it far*. The sweep then produces a nearby, much sparser set.

11.4 Why this is not spectral sparsification

A multiplicative spectral sparsifier H of G satisfies

$$(1-\eta)L_G \preceq L_H \preceq (1+\eta)L_G,$$

thus approximately preserving every quadratic form [6]. Such an approximation preserves the kernel. Kun repair, by contrast, intentionally disconnects the graph and therefore enlarges the kernel. So the operation is not spectral sparsification.

The conceptual overlap is instead that both theories manipulate the Laplacian as a global quadratic-form object. In sparsification one preserves that object multiplicatively; here one exploits a local spectral inequality to identify an almost block-diagonal decomposition and then edits away the small off-block part.

12 The proof in one display

The entire argument can be summarized as

$$\begin{array}{c}
\text{local SOS validity} \\
\Downarrow \\
L^2 \succeq \gamma L - M_b, \qquad \mathbb{E}b = O(\varepsilon) \\
\Downarrow \quad Q_t = (I+tL)^{-1} \\
h\text{-sparse } T \text{ is close to a function of energy } O\left(\frac{|T|}{\gamma t^2}\right) \\
\Downarrow \quad \text{coarea / threshold sweep} \\
T \text{ is close to } U \text{ with } \Phi(U) = O\left(\frac{1}{t\sqrt{\gamma}}\right) \\
\Downarrow \quad \text{minimum-set peeling} \\
\text{near-expanders with } O\left(\varepsilon t^2 + \frac{1}{t\sqrt{\gamma}}\right) n \text{ crossing mass} \\
\Downarrow \quad \text{trimming + label completion} \\
\text{induced } S\text{-labelled components of conductance } \Omega(\gamma) \\
\Downarrow \quad t \asymp \varepsilon^{-1/3}\gamma^{-1/6} \\
\delta(\varepsilon) = O\left(\gamma^{-4/3}\varepsilon^{1/3}\right), \qquad \lambda_1 = \Omega(\gamma^2).
\end{array}$$

The main conceptual sentence is:

> *An almost-everywhere Kazhdan spectral certificate forces every genuinely sparse cut to be close to an arbitrarily sparser cut; minimum-set peeling turns that sharpening phenomenon into an expander decomposition.*

References

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